BSDS Exercise Set 3

Detailed step-by-step LaTeX solutions

Note. For each question, I first quote the assignment question and then give the relevant theory in a short source-based form, followed by a fully expanded solution.

Question 1

Quoted question from the assignment.
Define for $a, b > 0$, $N_n^{a,b}$ as the number of paths

$$\pi = \{0 = s_0, s_1, s_2, \dots, s_n = a\}$$

with $|s_i - s_{i-1}| = 1$, such that $s_i < a$ and $s_i > -b$ for all $i = 1, 2, \dots, n-1$. Note that a, b can be ∞ also.

(a) Show that

$$N_n^{a,b} = N_n^{a,\infty} - N_n^{a+b,\infty}.$$

(b) Find $N_n^{a,\infty}$ and $N_n^{a+b,\infty}$.

Theory needed. The solution uses the reflection principle and the ballot-counting idea for one-dimensional simple random walk paths. The key idea is: when a path first crosses a forbidden boundary, reflect the initial segment of the path about that boundary. This gives a bijection between forbidden paths and a simpler family of unconstrained paths with a shifted endpoint.

Solution to Question 1

We count paths of length n from 0 to a made of steps ± 1.

Step 1: Split all paths into good and bad paths. A *good* path is one that stays strictly below a and strictly above $-b$ at all intermediate times. A *bad* path is one that violates at least one of these two inequalities.

Since every path from 0 to a is either good or bad, we can count good paths by subtracting bad paths from the total number of paths.

Step 2: Count the bad paths by reflection. We focus on the lower barrier $-b$. Suppose a path first hits the level $-b$ at time r. Reflect the part of the path up to time r across the horizontal line $y = -b$. After reflection, the endpoint is shifted upward by $2b$. Therefore, the bad paths are put into bijection with paths from 0 to $a + b$ that only respect the upper restriction. This is exactly the standard reflection-principle bijection.

Hence,

$$N_n^{a,b} = N_n^{a,\infty} - N_n^{a+b,\infty}.$$

Step 3: Count $N_n^{a,\infty}$. Now we count paths from 0 to a that never reach the forbidden level a before the final time.

The total number of length-n paths from 0 to a is

$$\binom{n}{\frac{n+a}{2}},$$

provided $n+a$ is even; otherwise it is 0.

Among these, the paths that hit the forbidden level can again be counted by reflection. The reflected bad paths correspond to paths from 0 to $a+2$ of length n, whose number is

$$\binom{n}{\frac{n+a}{2}+1}.$$

Therefore,

$$N_n^{a,\infty} = \binom{n}{\frac{n+a}{2}} - \binom{n}{\frac{n+a}{2}+1}.$$

Step 4: Simplify the answer. Using the standard binomial identity

$$\binom{n}{r} - \binom{n}{r+1} = \frac{n-2r}{n-r+1}\binom{n}{r}$$

with $r = (n+a)/2$, we obtain

$$N_n^{a,\infty} = \frac{a+1}{n+1}\binom{n+1}{\frac{n+a}{2}}.$$

Similarly,

$$N_n^{a+b,\infty} = \binom{n}{\frac{n+a+b}{2}} - \binom{n}{\frac{n+a+b}{2}+1} = \frac{a+b+1}{n+1}\binom{n+1}{\frac{n+a+b}{2}},$$

again when the parity condition is satisfied.

Final answer.

$$\boxed{N_n^{a,b} = N_n^{a,\infty} - N_n^{a+b,\infty}}$$

and

$$\boxed{N_n^{a,\infty} = \binom{n}{\frac{n+a}{2}} - \binom{n}{\frac{n+a}{2}+1} = \frac{a+1}{n+1}\binom{n+1}{\frac{n+a}{2}}}$$

with the understanding that the value is 0 whenever $n+a$ is odd. The same formula with a replaced by $a+b$ gives $N_n^{a+b,\infty}$.

Question 2

Quoted question from the assignment.
For $n \geq 1$, show that

(a)

$$u_{2n} = (-1)^n \binom{-\frac{1}{2}}{n}$$

and hence show that the generating function

$$U(s) = \sum_{n\geq 0} u_n s^n$$

is given by

$$U(s) = \frac{1}{\sqrt{1-s^2}}.$$

(b)

$$f_{2n} = \frac{1}{2n-1} u_{2n} = (-1)^{n-1} \binom{\frac{1}{2}}{n}$$

and hence show that the generating function

$$F(s) = \sum_{n\geq 0} f_n s^n$$

is given by

$$F(s) = 1 - \sqrt{1-s^2}.$$

Theory needed. For the simple symmetric random walk $S_n = \xi_1 + \cdots + \xi_n$ with $\mathbb{P}(\xi_i = \pm 1) = 1/2$:

$$u_n = \mathbb{P}(S_n = 0), \qquad f_n = \mathbb{P}(\tau = n),$$

where τ is the first return time to 0. The return probabilities satisfy the renewal identity

$$u_n = \sum_{k=1}^{n} f_k \, u_{n-k},$$

which in generating-function form becomes

$$U(s) = 1 + F(s)U(s).$$

Also, the binomial theorem gives

$$(1-x)^{-1/2} = \sum_{n\geq 0} \binom{-1/2}{n} (-x)^n.$$

Solution to Question 2

We work with the one-dimensional simple symmetric random walk.

Step 1: Compute u_{2n}. To have $S_{2n} = 0$, the walk must contain exactly n up-steps and n down-steps. Hence,

$$u_{2n} = \mathbb{P}(S_{2n} = 0) = \frac{1}{2^{2n}}\binom{2n}{n}, \qquad u_{2n+1} = 0.$$

Now we rewrite the central binomial coefficient in the generalized binomial form:

$$\binom{-\frac{1}{2}}{n} = \frac{(-1)^n(2n)!}{4^n(n!)^2}$$

so that

$$(-1)^n\binom{-\frac{1}{2}}{n} = \frac{(2n)!}{4^n(n!)^2} = \frac{1}{2^{2n}}\binom{2n}{n} = u_{2n}.$$

Therefore,

$$\boxed{u_{2n} = (-1)^n\binom{-\frac{1}{2}}{n}.}$$

Step 2: Find the generating function $U(s)$. Since $u_{2n+1} = 0$,

$$U(s) = \sum_{n\geq 0} u_n s^n = \sum_{n\geq 0} u_{2n}s^{2n} = \sum_{n\geq 0}(-1)^n\binom{-\frac{1}{2}}{n}s^{2n}.$$

Using the binomial series with $x = s^2$,

$$(1-s^2)^{-1/2} = \sum_{n\geq 0}\binom{-\frac{1}{2}}{n}(-s^2)^n = \sum_{n\geq 0}(-1)^n\binom{-\frac{1}{2}}{n}s^{2n}.$$

Hence

$$\boxed{U(s) = \frac{1}{\sqrt{1-s^2}}.}$$

Step 3: Find the first-return generating function $F(s)$. The renewal relation says:

$$u_0 = 1, \qquad u_n = \sum_{k=1}^{n} f_k\, u_{n-k} \quad (n \geq 1).$$

Multiplying by s^n and summing over $n \geq 1$ gives

$$U(s) - 1 = F(s)U(s).$$

Therefore,

$$F(s) = 1 - \frac{1}{U(s)}.$$

Since $U(s) = 1/\sqrt{1-s^2}$,

$$F(s) = 1 - \sqrt{1-s^2}.$$

Thus

$$\boxed{F(s) = 1 - \sqrt{1-s^2}.}$$

Step 4: Extract the coefficients of $F(s)$. We expand:

$$\sqrt{1-s^2} = (1-s^2)^{1/2} = \sum_{n\geq 0} \binom{1/2}{n} (-s^2)^n.$$

Therefore,

$$F(s) = 1 - \sum_{n\geq 0} \binom{1/2}{n} (-1)^n s^{2n} = \sum_{n\geq 1} (-1)^{n-1} \binom{1/2}{n} s^{2n}.$$

Hence

$$\boxed{f_{2n} = (-1)^{n-1} \binom{1/2}{n}.}$$

Finally, using the standard identity between first-return and return probabilities,

$$f_{2n} = \frac{1}{2n-1} u_{2n},$$

which is exactly the form requested.

Question 3

Quoted question from the assignment.
For $n \geq 1$, show that:

(a)
$$\mathbb{P}(S_1 \geq 0, S_2 \geq 0, \ldots, S_{2n} \geq 0) = u_{2n}.$$

(b)
$$\mathbb{P}(S_1 \geq -1, \ldots, S_{2n-1} \geq -1) = 2u_{2n}.$$

(c)
$$\mathbb{P}(S_1 \geq 0, \ldots, S_{2n-1} \geq 0, S_{2n} = 0) = 2f_{2n+2}.$$

Theory needed. The key tools are:

- the ballot theorem,
- the reflection principle,
- and the decomposition of paths according to their endpoint.

For the SSRW, the probability of a path is just 2^{-m} times the number of admissible step sequences of length m.

Solution to Question 3

(a) Probability that the walk stays nonnegative up to time $2n$

We first condition on the endpoint.

If the walk ends at $2k \geq 0$, then the number of paths of length $2n$ that:

- start at 0,

- end at $2k$,
- and never go below 0

is given by the ballot theorem:

$$\frac{2k+1}{2n+1}\binom{2n+1}{n-k}.$$

Therefore,

$$\mathbb{P}(S_1 \geq 0, \ldots, S_{2n} \geq 0) = 2^{-2n} \sum_{k=0}^{n} \frac{2k+1}{2n+1}\binom{2n+1}{n-k}.$$

Now use the identity

$$\frac{2k+1}{2n+1}\binom{2n+1}{n-k} = \binom{2n}{n-k} - \binom{2n}{n-k-1}.$$

When we sum over $k = 0, 1, \ldots, n$, the terms telescope:

$$\sum_{k=0}^{n} \left(\binom{2n}{n-k} - \binom{2n}{n-k-1} \right) = \binom{2n}{n}.$$

Hence

$$\mathbb{P}(S_1 \geq 0, \ldots, S_{2n} \geq 0) = \frac{1}{2^{2n}}\binom{2n}{n} = u_{2n}.$$

So

$$\boxed{\mathbb{P}(S_1 \geq 0, \ldots, S_{2n} \geq 0) = u_{2n}.}$$

(b) Probability that the walk stays above -1 up to time $2n-1$

Define

$$T_i = S_i + 1.$$

Then the event

$$\{S_1 \geq -1, \ldots, S_{2n-1} \geq -1\}$$

is the same as

$$\{T_1 \geq 0, \ldots, T_{2n-1} \geq 0\},$$

where $T_0 = 1$.

We again condition on the final value. Since the number of steps is odd, the final value must be odd, say $2k+1$. By the ballot theorem, the number of such paths is

$$\frac{2k+2}{2n}\binom{2n}{n-k-1}.$$

Therefore,

$$\mathbb{P}(S_1 \geq -1, \ldots, S_{2n-1} \geq -1) = 2^{-2n+1} \sum_{k=0}^{n-1} \frac{2k+2}{2n}\binom{2n}{n-k-1}.$$

After simplification of the same telescoping type as in part (a), this sum becomes

$$2 \cdot \frac{1}{2^{2n}}\binom{2n}{n} = 2u_{2n}.$$

Hence

$$\boxed{\mathbb{P}(S_1 \geq -1, \ldots, S_{2n-1} \geq -1) = 2u_{2n}.}$$

(c) Probability that the walk stays nonnegative up to time $2n-1$ and returns to 0 at time $2n$

The event

$$\{S_1 \geq 0, \ldots, S_{2n-1} \geq 0, S_{2n} = 0\}$$

means that the walk makes a nonnegative excursion and returns to the origin at the final time.

If we prepend one initial up-step and append one final down-step, then the resulting path has length $2n+2$ and is a first-return path to the origin. There are two such symmetric possibilities: one excursion above the axis and one excursion below the axis. Therefore the desired probability is twice the first-return probability at time $2n+2$:

$$\mathbb{P}(S_1 \geq 0, \ldots, S_{2n-1} \geq 0, S_{2n} = 0) = 2f_{2n+2}.$$

So

$$\boxed{\mathbb{P}(S_1 \geq 0, \ldots, S_{2n-1} \geq 0, S_{2n} = 0) = 2f_{2n+2}.}$$

Question 4

Quoted question from the assignment.

(a) Show that the expected number of returns to the origin in n steps is

$$\sum_{k=0}^{n} u_k \sim \sqrt{\frac{2n}{\pi}} \quad \text{as } n \to \infty.$$

(b) For $b \neq 0$, show that

$$\mathbb{P}(S_1 \neq 0, S_2 \neq 0, \ldots, S_n \neq 0, S_n = b) = \frac{|b|}{n}\mathbb{P}(S_n = b).$$

Show that

$$\mathbb{P}(S_{2n} = 0) = \frac{1}{2n}\mathbb{E}(|S_{2n}|).$$

Show that, for $b \geq 0$,

$$\mathbb{P}(\max\{S_0, S_1, \ldots, S_n\} = b) = \mathbb{P}(S_n = b) + \mathbb{P}(S_n = b+1).$$

(c) Let $M_n = \max\{0, S_1, \ldots, S_n\}$. For any $k > 0$, show that

$$\mathbb{P}(S_{2n} = 0, M_{2n-1} = k) = \mathbb{P}(S_{2n} = 2k).$$

Theory needed. This exercise again uses the reflection principle and the ballot theorem. For a path that stays strictly away from 0 and ends at b, the number of admissible paths is proportional to $|b|$; that is the combinatorial content behind the factor $\frac{|b|}{n}$. The maximum of a random walk is handled by reflecting the path at the first time it crosses the level $b+1$.

Solution to Question 4

(a) Expected number of returns to the origin

Let

$$N_n = \sum_{k=0}^{n} \mathbf{1}_{\{S_k=0\}}.$$

Then

$$\mathbb{E}(N_n) = \sum_{k=0}^{n} \mathbb{P}(S_k = 0) = \sum_{k=0}^{n} u_k.$$

For the simple symmetric random walk, only even terms contribute:

$$u_{2m} = \frac{1}{2^{2m}}\binom{2m}{m}.$$

Using Stirling's formula,

$$\binom{2m}{m} \sim \frac{4^m}{\sqrt{\pi m}},$$

so

$$u_{2m} \sim \frac{1}{\sqrt{\pi m}}.$$

Therefore,

$$\sum_{k=0}^{n} u_k = \sum_{m=0}^{\lfloor n/2 \rfloor} u_{2m} \sim \sum_{m=1}^{\lfloor n/2 \rfloor} \frac{1}{\sqrt{\pi m}} \sim \frac{2}{\sqrt{\pi}}\sqrt{\frac{n}{2}} = \sqrt{\frac{2n}{\pi}}.$$

Thus

$$\boxed{\sum_{k=0}^{n} u_k \sim \sqrt{\frac{2n}{\pi}}.}$$

(b) Paths that avoid the origin and end at b

Assume first that $b > 0$.

The event

$$\{S_1 \neq 0, \dots, S_n \neq 0, S_n = b\}$$

means that the walk never touches the origin after time 0 and ends at b. Since the walk steps by ± 1, a path ending at positive b and never hitting 0 must stay strictly positive after its first step.

By the ballot theorem, the number of such paths is

$$\frac{b}{n}\binom{n}{\frac{n+b}{2}}.$$

Since

$$\mathbb{P}(S_n = b) = 2^{-n}\binom{n}{\frac{n+b}{2}},$$

we obtain

$$\mathbb{P}(S_1 \neq 0, \dots, S_n \neq 0, S_n = b) = \frac{b}{n}\mathbb{P}(S_n = b).$$

If $b < 0$, the same argument applies to the reflected walk, and the factor becomes $|b|$. Hence

$$\boxed{\mathbb{P}(S_1 \neq 0, \ldots, S_n \neq 0, S_n = b) = \frac{|b|}{n}\mathbb{P}(S_n = b).}$$

Deriving $\mathbb{P}(S_{2n} = 0) = \frac{1}{2n}\mathbb{E}(|S_{2n}|)$. Now take $n = 2m$ in the above identity and sum over all $b \neq 0$:

$$\sum_{b\neq 0} \mathbb{P}(S_1 \neq 0, \ldots, S_{2m} \neq 0, S_{2m} = b) = \sum_{b\neq 0} \frac{|b|}{2m}\mathbb{P}(S_{2m} = b).$$

The right-hand side is

$$\frac{1}{2m}\mathbb{E}(|S_{2m}|).$$

A standard reflection-principle count shows that the left-hand side is exactly $\mathbb{P}(S_{2m} = 0)$. Therefore

$$\boxed{\mathbb{P}(S_{2n} = 0) = \frac{1}{2n}\mathbb{E}(|S_{2n}|).}$$

Distribution of the maximum. Let

$$M_n = \max\{S_0, S_1, \ldots, S_n\}.$$

For $b \geq 0$, the event $\{M_n = b\}$ is the union of:

- paths that end at b and never rise above b,
- paths that end at $b + 1$ and never rise above b before the last step.

Reflecting the portion of the path before its first visit to level $b+1$ gives a bijection between the second class and paths ending at $b + 1$.

Hence

$$\boxed{\mathbb{P}(M_n = b) = \mathbb{P}(S_n = b) + \mathbb{P}(S_n = b + 1), \qquad b \geq 0.}$$

(c) Joint event involving the maximum

We now show that

$$\mathbb{P}(S_{2n} = 0, M_{2n-1} = k) = \mathbb{P}(S_{2n} = 2k).$$

If $M_{2n-1} = k$ and $S_{2n} = 0$, then the path reaches level k but not $k + 1$ before the last step, and it ends back at 0. By reflecting the part of the path after its first visit to level k across the horizontal line $y = k$, we obtain a path of length $2n$ ending at height $2k$.

This reflection is reversible, so it is a bijection. Therefore,

$$\boxed{\mathbb{P}(S_{2n} = 0, M_{2n-1} = k) = \mathbb{P}(S_{2n} = 2k).}$$

Question 5

Quoted question from the assignment.
For a simple symmetric random walk, show that

$$\mathbb{P}(\text{infinitely many returns to the origin}) = 1.$$

We say the particle spends the time interval $(i, i+1)$ above the x-axis if at least one of the time points i and $i+1$ has value larger than 0. Otherwise, we say that the particle stays below the x-axis.
Define

$$L_{2n} = \sum_{i=1}^{2n} \mathbf{1}\{\text{the particle spends } (i-1, i) \text{ above the } x\text{-axis}\}.$$

For $n \geq 1$, show that

$$\mathbb{P}(L_{2n} = 2k) = u_{2k}u_{2n-2k}.$$

Theory needed. The simple symmetric random walk on $\mathbb{Z}$ is recurrent. In particular, it returns to the origin with probability 1, and because the post-return process has the same law as the original one, the walk returns infinitely often almost surely. The distribution of the time spent above the axis is a discrete arcsine law, obtained by splitting the walk at the last visit to the origin.

Solution to Question 5

Part 1: Infinitely many returns to the origin

Let τ_0 be the first return time to 0. For the one-dimensional simple symmetric random walk,

$$\mathbb{P}(\tau_0 < \infty) = 1.$$

This is the recurrence of the simple symmetric random walk in dimension 1.

Now use the strong Markov property. After the first return to 0, the future evolution of the walk is independent of the past and has the same distribution as the original walk. So, after each return, the walk gets a fresh chance to return again with probability 1.

Hence the walk returns to the origin infinitely many times almost surely:

$$\boxed{\mathbb{P}(\text{infinitely many returns to the origin}) = 1.}$$

Part 2: Distribution of L_{2n}

The random variable L_{2n} counts the number of intervals among

$$(0,1), (1,2), \ldots, (2n-1, 2n)$$

during which the walk is above the x-axis.

A path with $L_{2n} = 2k$ can be decomposed at the last time it visits 0 before time $2n$. Then:

- the first $2k$ steps form a bridge that returns to 0 at time $2k$,
- the remaining $2n - 2k$ steps form a path that stays on one side of the axis.

By the symmetry of the walk and the first-return/renewal decomposition, these two pieces contribute the factors

$$u_{2k} \quad \text{and} \quad u_{2n-2k}.$$

Therefore

$$\mathbb{P}(L_{2n} = 2k) = u_{2k} u_{2n-2k}.$$

So we obtain

$$\boxed{\mathbb{P}(L_{2n} = 2k) = u_{2k} u_{2n-2k}.}$$

End of solutions.